4 Eigenvalues and eigenvectors

4.1 Eigenvectors and eigenvalues

Many of the uses and applications of linear algebra are especially evident by considering diagonal matrices. In addition to the fact that they are easy to multiply, a number of other properties readily emerge: their determinants (see Section 1.5) are trivial to compute, we can quickly determine whether such matrices have inverses, and, when they do, their inverses are easy to obtain. Thus, diagonal matrices are simple matrix representations for linear transformations from a finite-dimensional vector space $\mathbb{V}$ to itself (see Section 3.4). Unfortunately, not all linear transformations from $\mathbb{V}$ to $\mathbb{V}$ can be represented by diagonal matrices. In this section and Section 4.3, we determine which linear transformations have diagonal matrix representations and which bases generate those representations.

To gain insight into the conditions needed to produce a diagonal matrix representation, we consider a linear transformation $\boldsymbol{T}$: $\mathbb{R}^3 \to \mathbb{R}^3$ having the diagonal matrix representation

$$\mathbf{D} = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

with respect to the basis $\mathbb{B} = \{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$. The first column of $\mathbf{D}$ is the coordinate representation of $\boldsymbol{T}(\mathbf{x}_1)$ with respect to $\mathbb{B}$, the second column of $\mathbf{D}$ is the coordinate representation of $\boldsymbol{T}(\mathbf{x}_2)$ with respect to $\mathbb{B}$, and the third column of $\mathbf{D}$ is the coordinate representation of $\boldsymbol{T}(\mathbf{x}_3)$ with respect to $\mathbb{B}$. That is,

$$T(\mathbf{x}_1) = \lambda_1\mathbf{x}_1 + 0\mathbf{x}_2 + 0\mathbf{x}_3 = \lambda_1\mathbf{x}_1$$

$$T(\mathbf{x}_2) = 0\mathbf{x}_1 + \lambda_2\mathbf{x}_2 + 0\mathbf{x}_3 = \lambda_2\mathbf{x}_2$$

$$T(\mathbf{x}_3) = 0\mathbf{x}_1 + 0\mathbf{x}_2 + \lambda_3\mathbf{x}_3 = \lambda_3\mathbf{x}_3$$

Mapping the basis vectors $\mathbf{x}_1$, $\mathbf{x}_2$, or $\mathbf{x}_3$ from the domain of $\boldsymbol{T}$ to the range of $\boldsymbol{T}$ is equivalent to simply multiplying each vector by the scalar λ_1, λ_2, or λ_3, respectively.

We say that a nonzero vector $\mathbf{x}$ is an *eigenvector* of a linear transformation $\boldsymbol{T}$ if there exists a scalar λ such that

$$\boldsymbol{T}(\mathbf{x}) = \lambda\mathbf{x}. \tag{4.1}$$

Linear Algebra. https://doi.org/10.1016/B978-0-12-823470-9.00004-5

In terms of a matrix representation $\mathbf{A}$ for $\boldsymbol{T}$, we define a nonzero vector $\mathbf{x}$ to be an eigenvector of $\mathbf{A}$ if there exists a scalar λ such that

$$\mathbf{Ax} = \lambda\mathbf{x} \tag{4.2}$$

A nonzero vector $\mathbf{x}$ is an eigenvector of a square matrix $\mathbf{A}$ if there exists a scalar λ, called an eigenvalue, such that $\mathbf{Ax} = \lambda\mathbf{x}$.

The scalar λ in equation (4.1) is an *eigenvalue* of the linear transformation $\boldsymbol{T}$; the scalar λ in equation (4.2) is an eigenvalue of the matrix $\mathbf{A}$. Note that an eigenvector must be nonzero; eigenvalues, however, may be zero.

Eigenvalues and eigenvectors have an interesting geometric interpretation in $\mathbb{R}^2$ or $\mathbb{R}^3$ when the eigenvalues are real. As described in Section 2.1, multiplying a vector in either vector space by a real number λ results in an elongation of the vector by a factor of $|\lambda|$ when $|\lambda| > 1$ or a contraction of the vector by a factor of $|\lambda|$ when $|\lambda| < 1$, followed by no rotation when λ is positive or a rotation of 180° when λ is negative. These four possibilities are illustrated in Figure 4.1 for the vector $\mathbf{u}$ in $\mathbb{R}^2$ with $\lambda = 1/2$ and $\lambda = -1/2$ and for the vector $\mathbf{v}$ in $\mathbb{R}^2$ with $\lambda = 3$ and $\lambda = -2$. Thus, an eigenvector $\mathbf{x}$ of a linear transformation $\boldsymbol{T}$ in $\mathbb{R}^2$ or $\mathbb{R}^3$ is always mapped into a vector $\boldsymbol{T}(\mathbf{x})$ that is parallel to $\mathbf{x}$.

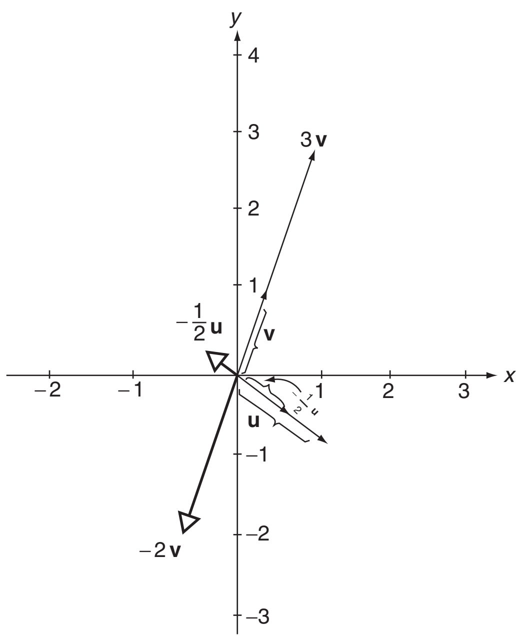

FIGURE 4.1

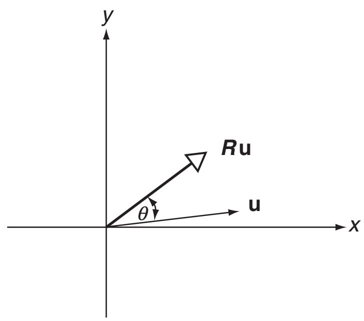

FIGURE 4.2

Not every linear transformation has real eigenvalues. Under the rotation transformation $\boldsymbol{R}$ described in Example 7 of Section 3.2, each vector is rotated around the origin by an angle θ in the counterclockwise direction (see Figure 4.2). As long as θ is not an integral multiple of 180°, *no* nonzero vector is mapped into another vector parallel to itself.

Example 1 The vector $\mathbf{x} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ is an eigenvector of $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$ because

$$\mathbf{Ax} = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}\begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = (-1)\begin{bmatrix} -1 \\ 1 \end{bmatrix} = (-1)\mathbf{x}$$

The corresponding eigenvalue is $\lambda = -1$.

Example 2 The vector $\mathbf{x} = \begin{bmatrix} 4 \\ 1 \\ -2 \end{bmatrix}$ is an eigenvector of $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ because

$$\mathbf{Ax} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}\begin{bmatrix} 4 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = 0\begin{bmatrix} 4 \\ 1 \\ -2 \end{bmatrix} = 0\mathbf{x}.$$

The corresponding eigenvalue is $\lambda = 0$.

Eigenvectors and eigenvalues come in pairs. If $\mathbf{x}$ is an eigenvector of a matrix $\mathbf{A}$, then there must exist an eigenvalue λ such that $\mathbf{Ax} = \lambda\mathbf{x}$, which is equivalent to the equation $\mathbf{Ax} - \lambda\mathbf{x} = \mathbf{0}$ or

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}. \tag{4.3}$$

Note that we cannot write equation (4.3) as $(\mathbf{A} - \lambda)\mathbf{x} = \mathbf{0}$ because subtraction between a scalar λ and a matrix $\mathbf{A}$ is undefined. In contrast, $\mathbf{A} - \lambda\mathbf{I}$ is the difference between two matrices, which is defined when $\mathbf{A}$ and $\mathbf{I}$ have the same order.

Equation (4.3) is a linear homogeneous equation for the vector **x**. If $(\mathbf{A} - \lambda\mathbf{I})^{-1}$ exists, we can solve equation (4.3) for **x**, obtaining $\mathbf{x} = (\mathbf{A} - \lambda\mathbf{I})^{-1}\mathbf{0} = \mathbf{0}$, which violates the condition that an eigenvector be nonzero. It follows that **x** is an eigenvector for **A** corresponding to the eigenvalue λ if and only if $(\mathbf{A} - \lambda\mathbf{I})$ does *not* have an inverse. Alternatively, because a square matrix has an inverse if and only if its determinant is nonzero, we may conclude that **x** is an eigenvector for **A** corresponding to the eigenvalue λ if and only if

To find eigenvalues and eigenvectors for a matrix **A**, first solve the characteristic equation, equation (4.4), for the eigenvalues and then for each eigenvalue solve equation (4.3) for the corresponding eigenvectors.

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0. \tag{4.4}$$

Equation (4.4) is the *characteristic equation* of **A**. If **A** has order $n \times n$, then the expansion of the left side of the equation, $\det(\mathbf{A} - \lambda\mathbf{I})$, results in an nth degree polynomial in λ called the *characteristic polynomial* of **A**. The characteristic equation of **A** has exactly n roots, which are the eigenvalues of **A**. Once an eigenvalue is located, corresponding eigenvectors are obtained by solving equation (4.3).

Example 3 Find the characteristic polynomial, eigenvalues, and eigenvectors of

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}.$$

Solution:

$$\mathbf{A} - \lambda\mathbf{I} = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-\lambda & 2 \\ 4 & 3-\lambda \end{bmatrix}.$$

The characteristic polynomial of **A** is $\det(\mathbf{A} - \lambda\mathbf{I}) = (1-\lambda)(3-\lambda) - 8 = \lambda^2 - 4\lambda - 5$. The characteristic equation of **A** is $\lambda^2 - 4\lambda - 5 = 0$, having as its roots $\lambda = -1$ and $\lambda = 5$. These two roots are the eigenvalues of **A**.

Eigenvectors of **A** have the form $\mathbf{x} = [x\ y]^{\mathrm{T}}$. With $\lambda = -1$, equation (4.3) becomes

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \left\{ \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} - (-1) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

or

$$\begin{bmatrix} 2 & 2 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The solution to this homogeneous matrix equation is $x = -y$, with y arbitrary. The eigenvectors corresponding to $\lambda = -1$ are

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ y \end{bmatrix} = y\begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

for any nonzero scalar y. We restrict y to be nonzero to ensure that the eigenvectors are nonzero.

With $\lambda = 5$, equation (4.3) becomes

$$(\mathbf{A} - \lambda\mathbf{I}) = \left\{\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} - 5\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right\}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

or

$$\begin{bmatrix} -4 & 2 \\ 4 & -2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The solution to this homogeneous matrix equation is $x = y/2$, with y arbitrary. The eigenvectors corresponding to $\lambda = 5$ are

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y/2 \\ y \end{bmatrix} = \frac{y}{2}\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

for any nonzero scalar y.

Example 4 Find the characteristic polynomial, eigenvalues, and eigenvectors of

$$\mathbf{A} = \begin{bmatrix} 2 & -1 & 0 \\ 3 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Solution:

$$\mathbf{A} - \lambda\mathbf{I} = \begin{bmatrix} 2 & -1 & 0 \\ 3 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \lambda\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2-\lambda & -1 & 0 \\ 3 & -2-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{bmatrix}.$$

Using expansion by cofactors with the last row, we find that the characteristic polynomial of $\mathbf{A}$ is

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (1-\lambda)[(2-\lambda)(-2-\lambda) + 3] = (1-\lambda)\left(\lambda^2 - 1\right).$$

The characteristic equation of **A** is $(1 - \lambda)(\lambda^2 - 1) = 0$; hence the eigenvalues of **A** are $\lambda_1 = \lambda_2 = 1$ and $\lambda_3 = -1$.

Eigenvectors of **A** have the form $\mathbf{x} = [x\ y\ z]^{\mathrm{T}}$. With $\lambda = 1$, equation (4.3) becomes

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \left\{ \begin{bmatrix} 2 & -1 & 0 \\ 3 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix} - (1) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

or

$$\begin{bmatrix} 1 & -1 & 0 \\ 3 & -3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The solution to this homogeneous matrix equation is $x = y$, with both y and z arbitrary. The eigenvectors corresponding to $\lambda = 1$ are

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} y \\ y \\ z \end{bmatrix} = y \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

for y and z arbitrary but not both zero to ensure that the eigenvectors are nonzero.

With $\lambda = -1$, equation (4.3) becomes

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \left\{ \begin{bmatrix} 2 & -1 & 0 \\ 3 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix} - (-1) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

or

$$\begin{bmatrix} 3 & -1 & 0 \\ 3 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The solution to this homogeneous matrix equation is $x = y/3$ and $z = 0$, with y arbitrary. The eigenvectors corresponding to $\lambda = -1$ are

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} y/3 \\ y \\ z \end{bmatrix} = \frac{y}{3} \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$$

for any nonzero scalar y.

The roots of a characteristic equation can be repeated. If $\lambda_1 = \lambda_2 = \lambda_3 = \ldots \lambda_k$, the eigenvalue is said to be of multiplicity k. Thus, in Example 4, $\lambda = 1$ is an eigenvalue of multiplicity 2, while $\lambda = -1$ is an eigenvalue of multiplicity 1.